Majorana minimal model exercices

by Francisco Lobo

Write the answers in *LyX* as formal and descriptive as possible. Leave intermediate steps you did but organize them outside the main text by putting them inside a colored box.

For this exercises it might be useful to consult:

1. Lutchyn, Roman M., Jay D. Sau, and S. Das Sarma. "Majorana fermions and a topological phase transition in semiconductor-superconductor heterostructures." Physical review letters 105.7 (2010): 077001.
2. Oreg, Yuval, Gil Refael, and Felix Von Oppen. "Helical liquids and Majorana bound states in quantum wires." Physical review letters 105.17 (2010): 177002.
3. Prada, Elsa, et al. "From Andreev to Majorana bound states in hybrid superconductor–semiconductor nanowires." Nature Reviews Physics 2.10 (2020): 575-594.
4. Chamon, C., et al. "Quantizing Majorana fermions in a superconductor." Physical Review B 81.22 (2010): 224515
5. Sau, Jay D., et al. "Generic new platform for topological quantum computation using semiconductor heterostructures." Physical review letters 104.4 (2010): 040502.

I. ANALITYCS

A. Types of superconductivity

Write a small essay on the basics of superconductivity BCS theory and make the distinction between s-wave and p-wave type superconductivity. Make comments on the meaning of the order parameter $\Delta(x) = \Delta_0(x)e^{i\varphi(x)}$ and symmetry considerations.

Answer:

B. Real space Hamiltonian

(B.1) Consider the system show in Fig.(1) being described by the Hamiltonian

$$\hat{\boldsymbol{H}} = \sum_{\sigma_i\sigma_j} c^\dagger_{i\sigma_i}\left(-\eta\partial_x^2 - \mu + \alpha_{\mathrm{R}} i\sigma_y\partial_x + V_{\mathrm{Z}}\sigma_z\right)c_{j\sigma_j} + \frac{1}{2}\left(\Delta_0 c^\dagger_{i\uparrow}c^\dagger_{j\downarrow} + \Delta_0 c_{i\downarrow}c_{j\uparrow}\right). \tag{1}$$

Identify each quantity on this expression and match each of the 5 terms on the Hamiltonian to its respective effect; kinetic, dopping, SOC, Zeeman and superconductivity. Describe each term, what generates them and invoke related useful formulas.

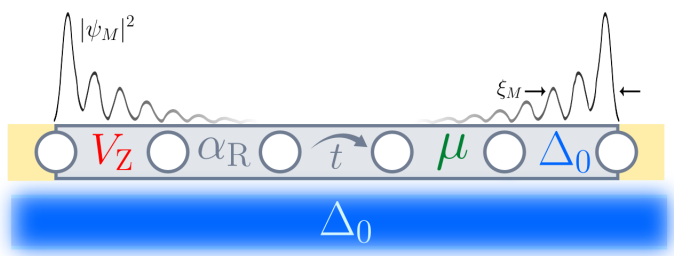

Figure 1. **(B.2)** Recreate this image on *Inkscape*.

Answer:

C. BdG Hamiltonian

(C.1) Express the Hamiltonian in Eq.(1) in it's Bogoliubov-de Genes (BdG) form reads as

$$\boldsymbol{H} = \frac{1}{2}\check{\boldsymbol{c}}_i^\dagger \boldsymbol{H}_{\mathrm{BdG}}\check{\boldsymbol{c}}_j = \begin{pmatrix} \boldsymbol{c}_i^\dagger & \boldsymbol{c}_i \end{pmatrix}\begin{pmatrix} \boldsymbol{h} & \boldsymbol{\Delta} \\ \boldsymbol{\Delta}^* & -\boldsymbol{h}^T \end{pmatrix}\begin{pmatrix} \boldsymbol{c}_j \\ \boldsymbol{c}_j^\dagger \end{pmatrix}, \tag{2}$$

with $\boldsymbol{h}$ hermitian and $\boldsymbol{\Delta}$ skew-symmetric, which is enforced from the fermionic statistics. **(C.2)** Prove this. Their explicit form read as

$$\boldsymbol{h} \equiv \begin{pmatrix} -\eta\partial_x^2 - \mu & V_{\mathrm{Z}} + \alpha_{\mathrm{R}}\partial_x \\ V_{\mathrm{Z}} - \alpha_{\mathrm{R}}\partial_x & -\eta\partial_x^2 - \mu \end{pmatrix} \text{ and } \boldsymbol{\Delta} \equiv \Delta_0\mathbb{1} \tag{3}$$

with $\check{\boldsymbol{c}}_\alpha$ are the Nambu eigen-vectors of the BdG Hamiltonian with real eigen-energies E_α, $\boldsymbol{H}_{\mathrm{BdG}}\check{\boldsymbol{c}}_\alpha = E_\alpha \check{\boldsymbol{c}}_\alpha$. Following Nambu's convection, we double the system's degrees of freedom and write the four-component Nambu spinor

$$\check{\boldsymbol{c}}_i = \begin{pmatrix} \boldsymbol{c}_i \\ \boldsymbol{c}_i^\dagger \end{pmatrix} = \begin{pmatrix} c_{i\uparrow} \\ c_{i\downarrow} \\ c_{i\uparrow}^\dagger \\ c_{i\downarrow}^\dagger \end{pmatrix} \tag{4}$$

Answer:

(C.2) Referring to both σ_i and τ_i as Pauli matrices operating only in their respectively spin or particle-hole space subspaces show that the BdG Hamiltonian can be expressed as

$$\boldsymbol{H} = (f\delta_{ij} - t\delta_{ij\pm1})\left[\tau_z \otimes \sigma_0\right] + V_{\mathrm{Z}}\left[\tau_z \otimes \sigma_z\right]\delta_{ij} + \frac{\alpha_{\mathrm{R}}}{2a_0}\delta_{ij\pm1}^{\pm}\left[\tau_z \otimes i\sigma_y\right] + \Delta_0\left[\tau_y \otimes \sigma_y\right] \tag{5}$$

with $f = 2t - \mu$ the band filling ($\mu = 0$ and $\mu = 2t$ corresponding to half and zero filling respectively), $t = \eta/a_0^2$ the hopping parameter and with the Kronecker delta δ_{ij} denoting the on-site terms and $\delta_{ij\pm1}^{\pm}$ the 1st neighbors hoppings (front and back). Write the complete explicit form of the matrix in Nambu space.

Remember that the discretized partial derivatives (in central differencing) read as

$$\partial_x c_i \to \frac{c_{i+1} - c_{i-1}}{2\delta_x} \equiv \frac{1}{2\delta_x}\begin{pmatrix} 0 & 1 & 0 & \dots \\ -1 & 0 & 1 & \dots \\ 0 & -1 & 0 & \ddots \\ \vdots & \vdots & \ddots & \ddots \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} \tag{6}$$

Answer:

D. Topological phase transition

(D.1) Consider the case of an infinite nanowire such that $\partial_x \to k_x \equiv k$. Using *Mathematica* solve the Hamiltonian's matrix characteristic equation and obtain the nanowire spectrum

$$E_\pm^2(k) = \left(\eta k^2 - \mu\right)^2 + \alpha_{\mathrm{R}}^2 k^2 + V_{\mathrm{Z}}^2 + \Delta_0^2 \pm 2\sqrt{\left(\eta k^2 - \mu\right)^2\left(\alpha_{\mathrm{R}}^2 k^2 + V_{\mathrm{Z}}^2\right) + V_{\mathrm{Z}}^2\Delta_0^2}, \tag{7}$$

(D.2) Why does the spectrum have four solution? **(D.3)** What happens at $k = 0$? Pinpoint the trivial-topological phase transition as a function of Zeeman. **(D.4)** What happens if one does not consider SOC?

Answer:

II. NUMERICS

This exercises are meant to be done in *Julia* using the *Quantica.jl* package. Consult to extensive documentation to know what to do.

To practice the workflow organize the code within *VScode* in three different folder: source code (*src*), simulations (*sim*) and analysis (*anal*). Within the *src* folder create a Minimal Majorana Model module using the necessary packages, exporting and including the necessary functions from the different *.jl* files. Make sure that each *sim.jl* file can be executable in *slurm batch*. Also, make the plots look polished using the *CairoMakie* package.

For the answers take prints of the code or find a pretty way to display the code in *LyX*.

A. Building the Hamiltonians

Build a parametric (as in depending on an input value of μ, V_{Z}, α and Δ_0) Hamiltonian as in Eq.(5) for both a finite and infinite 1D nanowire as in Fig.(1).

For this you will need to **1)** construct a linear lattice using *Quantica.LP.linear* (consider $R_{\mathrm{nw}} \lesssim k_F^{-1}$ such that there is only one 1D transversal mode), **2)** make a parametric model as a sum of each of the 5 different terms using *Quantica.onsite* and *Quantica.hopping* and **3)** generate the Hamiltonian of that model in the given lattice using *Quantica.hamiltonian*. Note that in the Nambu scheme there are four orbitals.

Answer:

B. Examine band structure for infinite case

Write a small essay on how the different terms effect the band structure with plot examples using the *Quantica.bands*. Comment on how the inner and outer gaps, often known as gap and minigap depending on each one is smaller, evolves as one tweaks the different parameters. What happens if one doesn't consider SOC?

For the solver use the *shifted invert Arnoldi method* and give a very quick explanation on how it works.

Answer:

C. Examine gap phase diagram

Calculate and display the gap phase diagram V_{Z} v.s. μ having Δ_0 fixed. Write a small essay on what you see and the differences between the finite and infinite case. Pinpoint the topological phase transition and check if the analytical solution coincides.

Answer:

D. Examine band spectrum for finite case

Calculate and display the band spectrum for the finite case as a function of V_{Z} having Δ_0 and μ fixed. Write a small essay on what you see referring to the topological phase transition and the Majorana oscillations.

Answer: